Useful Memory Has a Horizon

Structural Limits of Drift Tracking

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Abstract

Under drift, memory does not stay equally useful: retaining more past information reduces statistical noise, but also increases temporal misalignment with the current distribution. As the target law moves, each additional sample becomes useful and increasingly stale, so the problem is not simply how much to remember, but how long remembered evidence should remain operationally current.

We study this trade-off under Wasserstein nonstationarity. Cube-root scalings already appear in nonstationary optimization and bandit literature, but here the object is the finite-memory horizon itself. For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory and a staleness term that grows with drift, yielding a **cube-root law**: the **optimal memory horizon** scales as $\boldsymbol{n^* \asymp (C_K/\zeta)^{2/3}}$, while the corresponding **minimum tracking error** scales as $\boldsymbol{\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}}$. In the critical-window regime, we also prove a minimax lower bound, showing that this finite-memory floor is structural rather than estimator-specific.

Empirically, Gaussian drift experiments recover the predicted U-curve and cube-root scaling. Continuous-drift benchmarks expose a gap between temporal validity and changepoint evidence; alternating-timescales benchmarks show that contraction is easier than re-expansion; and gap-cost experiments show that horizon misalignment induces nonlinear predictive cost.

The resulting picture is operational: useful memory under drift has geometry, temporal coherence is not the same object as changepoint evidence, and the predictive value of horizon control depends on the interaction between drift geometry and memory policy.

Keywords: tracking under drift; useful memory; temporal coherence; Wasserstein distance; Sinkhorn divergence; concept drift; adaptive memory.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory must therefore solve a basic but easy-to-miss problem: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. Using more past data reduces sampling noise, but it also increases the age of the information being used to track the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The constraint is temporal rather than parametric: one cannot overparameterize time. The central claim of this paper is that this tension defines a coherence-lag trade-off and induces a quantitative law for finite-memory tracking under Wasserstein nonstationarity. The cube-root exponents themselves are not unique to this paper; what is new here is the finite-memory horizon interpretation and the temporal-validity floor.

For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory length and a staleness term that increases with drift and effective data age. Optimizing that balance yields a cube-root law: $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$ with $n^* \asymp (C_K/\zeta)^{2/3}$. The law is exact for the averaging class studied in the paper. It is not asserted as a theorem for every adaptive estimator. What the paper does show beyond that class is a minimax lower bound for window-restricted estimators in the critical-window regime, establishing that the finite-memory floor is structural rather than a quirk of one specific rule.

This viewpoint also gives the paper its interpretive layer. The limiting factor is the age of the information that finite memory forces the estimator to retain, alongside the usual finite-sample cost. The result can therefore be read as a temporal-coherence law: the learner pays for variance reduction by carrying older evidence, and optimal tracking occurs at a memory horizon that balances statistical stability against operational staleness.

The empirical story is more nuanced than a dominance claim. Some regimes reward a short fixed window as a strong operational compromise, even when the local oracle horizon moves over time; what the cube-root regulator adds is a way to track horizon validity, not a guarantee of lower error everywhere. This separates horizon geometry, horizon tracking, and predictive payoff.

The empirical section follows the same logic in four steps. Continuous-drift experiments recover the 'U'-curve and expose the gap between temporal validity and detection. The temporal-validity check measures the delay between cap activation and ADWIN detection, the alternating-timescales check exposes horizon instability, and the gap-cost analysis quantifies how misalignment translates into predictive cost. Sinkhorn-based tracking scores serve as the measurement layer, and the cube-root cap shows how the same law can act as an online regulator as well as a retrospective explanation.

The theoretical results are stated in W_2; the Sinkhorn divergence is the computational realization used in the synthetic measurement layer.

We also keep one representative public-stream case study in the main text, ELEC2, while the full ELEC2/Bikes arena is provided in appendix E as a robustness check rather than a leaderboard.

Within that last benchmark, we also isolate a secondary formal effect: an asymmetric horizon-recovery law for EMA-smoothed drift-to-horizon maps, where contraction after bursts is faster than re-expansion after low-drift stretches.

The window size is a fundamental parameter whose mischoice can create spurious or missed detections Hinder et al. [2023]; the coherence-lag law provides a principled answer when the drift rate is estimable.

The scope of the paper is deliberately narrow. We study finite-memory tracking under drift, a lower bound for restricted estimators, and the associated temporal-coherence interpretation, together with experiments chosen to test that story. The paper asks a single question: what tracking accuracy is possible when the learner's memory is finite and the target distribution keeps moving?

2 Useful Memory Geometry

Under continuous drift, memory is not only a statistical resource but also a temporal liability. The core question is therefore not whether one can react to change, but how long a sample should remain operationally useful before it becomes stale.

Theorem 2.1 (Lag-Inclusive Tracking Error Bound). *Let $X_{t-j} \sim P^*_{t-j}$ and let*

$$\widehat{P}_t^{(n)} \;=\; \frac{1}{n}\sum_{j=0}^{n-1} \delta_{X_{t-j}}$$

be the uniform-window empirical estimator. Assume the target path is locally W_2-Lipschitz with rate ζ, so that

$$W_2\big(P^*_{t-j}, P^*_t\big) \le \zeta j \qquad \text{for all } 0 \le j < n,$$

and assume the finite-sample term obeys

$$\mathbb{E}\Big[W_2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big)\Big] \le C_K n^{-1/2},$$

where $\bar{P}_t^{(n)} := n^{-1}\sum_{j=0}^{n-1} P^*_{t-j}$. Then the expected tracking error satisfies

$$\mathcal{E}(n) \;\le\; C_K\, n^{-1/2} \;+ \tfrac{1}{2}\,\zeta\, n,$$

where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.

Lemma 2.2 (Finite-sample Wasserstein term). *Let Q be a probability measure on $\mathbb{R}^{d_{\text{eff}}}$ with support contained in a ball of radius R and finite second moment. Let $X_1, \ldots, X_n$ be i.i.d. samples from Q, and let $\hat{Q}_n := n^{-1}\sum_{i=1}^n \delta_{X_i}$. Then:*

- *If $d_{\text{eff}} < 4$, there exists a constant $C_{d_{\text{eff}},R}$ such that*

$$\mathbb{E}\, W_2(\hat{Q}_n, Q) \le C_{d_{\text{eff}},R}\, n^{-1/2}.$$

- *If $d_{\text{eff}} = 4$, the same bound holds with an additional $(\log n)^{1/2}$ factor.*

- *If $d_{\text{eff}} > 4$, the raw Wasserstein rate is slower than $n^{-1/2}$, so the constant-only form above should not be used.*

If the geometry is computed with debiased Sinkhorn divergence S_ε on a bounded domain of diameter D, then for fixed $\varepsilon > 0$ the empirical fluctuations remain $O(n^{-1/2})$ with a constant $L(D, \varepsilon)$ that is dimension-free at fixed ε and increases as $\varepsilon \downarrow 0$.

A short numerical sanity check for this regime is reported in Appendix C; it is not used as a theorem, but it matches the intended low-dimensional interpretation of the finite-sample term.

The two terms in theorem 2.1 pull in opposite directions. Increasing n reduces variance but raises the age of the retained evidence. That tension is the basic coherence-lag trade-off that governs useful memory under drift.

Proposition 2.3 (Optimal Window Under Drift)**.** *Under theorem 2.1, the error $\mathcal{E}(n)$ is minimised at*

$$n^* = \left(\frac{C_K}{\zeta}\right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} = \tfrac{3}{2}\, C_K^{2/3}\, \zeta^{1/3}.$$

Proof. Apply the triangle inequality for W_2:

$$W_2\left(\widehat{P}_t^{(n)}, P_t^*\right) \le W_2\left(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\right) + W_2\left(\bar{P}_t^{(n)}, P_t^*\right).$$

The first term is controlled by the assumed finite-sample bound. For the second term, let π_j be an optimal coupling between P_{t-j}^* and P_t^*. Then the averaged coupling $\bar{\pi} := n^{-1}\sum_{j=0}^{n-1}\pi_j$ couples $\bar{P}_t^{(n)}$ with P_t^*, so the transport cost is at most the average of the individual costs. Combined with the W_2-Lipschitz drift bound, this gives

$$W_2\left(\bar{P}_t^{(n)}, P_t^*\right) \le \frac{1}{n}\sum_{j=0}^{n-1} W_2\left(P_{t-j}^*, P_t^*\right) \le \frac{1}{n}\sum_{j=0}^{n-1} \zeta j \le \tfrac{1}{2}\,\zeta n.$$

Combining the two bounds yields the stated inequality. □

Remark 2.4 (Measurement layer)*.* The paper's theorem is stated in W_2, so ε does not enter the mathematical drift law itself. In the synthetic measurement layer, however, the observable geometry is often a debiased Sinkhorn divergence with fixed ε, and then the effective finite-sample constant should be read as $C_K^{\text{eff}} = C_{d_{\text{eff}},R} + L(D, \varepsilon)$: the raw Wasserstein part depends on intrinsic dimension, while the Sinkhorn part is dimension-free at fixed ε but worsens as $\varepsilon \downarrow 0$.

In the synthetic benchmarks, $\hat{\zeta}_t$ is estimated from short-window block differences using an EMA update, C_K is calibrated from a stationary prefix, and ε is held fixed within each experiment family, with runtime sensitivity to ε reported separately. The benchmark-specific drift proxy, robustness check, and sensitivity sweep are detailed in Appendix D. The same appendix also validates an observable prequential calibration rule for choosing α and the block scale, so the online regulator is not tied to a fixed manual setting. For the horizon constant itself, the prefix calibration is the simple stationary baseline used to estimate the scale in the synthetic stream; it is not a tuned oracle, but it is also not arbitrary.

The Sinkhorn-to-W_2 gap is therefore treated as an empirical calibration constant. In the tested Gaussian shift sweep, the high-dimensional slice shows that this gap is nearly flat in sample size for small ε and decays with n only when ε is large; the resulting horizon inflation in the sampled grid ranges from about 1.36× to 4.24×. This supports the paper's use of fixed-ε Sinkhorn as a measurement layer while keeping the theoretical law in W_2. For fixed ε on bounded domains, the debiased Sinkhorn divergence converges at rate $n^{-1/2}$ with ε-dependent constants [Genevay et al., 2019]; the approximation gap to W_2^2 is $O(\varepsilon)$ and vanishes as $\varepsilon \to 0$. We therefore treat the Sinkhorn layer as a calibration proxy rather than part of the law itself.

Corollary 2.5 (EWMA Analogue). *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale.*

Definition 2.6 (Effective Memory Age). Let an estimator place weights $w_0, \ldots, w_{n-1}$ on the most recent observations, with $\sum_j w_j = 1$. Its *effective memory age* is

$$A(w) \;=\; \sum_{j=0}^{n-1} j\, w_j,$$

and its corresponding freshness proxy is the monotone transform $F(w) = 1/(1 + A(w))$. The role of $F(w)$ is purely geometric: it records how old the retained evidence is before any local drift scale is specified. For a uniform window, $A(w) = (n-1)/2$.

Definition 2.7 (Temporal Coherence). For an estimator with weights w at time t, define its *operational staleness*

$$S_t(w) \;=\; \sum_{j=0}^{n-1} w_j\, W_2(\mathcal{P}^*(S, t-j), \mathcal{P}^*(S, t))\,.$$

Its *temporal coherence* is the inverse alignment score

$$C_t(w) \;=\; \frac{1}{1 + S_t(w)}.$$

If the target path is locally W_2-Lipschitz at rate ζ_t, in the sense that $W_2(\mathcal{P}^*(S, t-j), \mathcal{P}^*(S, t)) \le \zeta_t j$ for the relevant lags, then

$$S_t(w) \;\le\; \zeta_t A(w), \qquad C_t(w) \;\ge\; \frac{1}{1 + \zeta_t A(w)}.$$

In that sense, $F(w)$ is the rate-free freshness proxy, while $C_t(w)$ measures operational alignment with the current distribution.

The quantity $A(w)$ gives a compact description of the temporal geometry of a finite-memory estimator. Short windows preserve temporal coherence at the cost of variance; long windows reduce variance at the cost of staleness. The cube-root horizon marks the point where that exchange stops improving. For a uniform window under approximately constant local drift, the induced staleness scales as $S_t(w) \approx \zeta(n-1)/2$, which is the same age term that appears in theorem 2.1 up to constants.

Proposition 2.8 (Finite-Memory Floor via Gaussian Location Subclass). *Let $\mathcal{P}_{\zeta,m}$ be the class of length-m distribution paths for which the following hold:*

1. *the path is W_2-Lipschitz with rate at most ζ;*
2. *the lower-bound construction is carried out on the Gaussian location subclass $X_t \sim \mathcal{N}(\mu_t, \sigma^2 I_d)$ with $\|\mu_t - \mu_s\|_2 \le \zeta|t-s|$;*
3. *the estimator at time m only uses the last m samples.*

Measure risk by the endpoint W_2 error, which coincides with Euclidean endpoint error in the Gaussian location subclass. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E}\, d\big(\hat{\mathcal{P}}_m, \mathcal{P}^*\big) \;\gtrsim\; \max\big\{C_K\, m^{-1/2},\; c\,\sigma^{2/3}\zeta^{1/3}\big\}\,.$$

Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural floor of finite memory on this drift class.

Proof. Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$. We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in the admissible drift class. For the two hypotheses $\theta_\pm = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\mathrm{KL}(P_+, P_-) = \frac{1}{2\sigma^2}\sum_{j=1}^{m}(\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}}\right)^2 = \frac{1}{2}.$$

For any estimator δ_m, the plug-in test $\psi = \mathbf{1}\{\delta_m \ge 0\}$ must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least $\sigma/(2\sqrt{m})$. By Le Cam's two-point lemma and Pinsker's inequality, the average testing error is at least

$$\frac{1 - \mathrm{TV}(P_+, P_-)}{2} \ge \frac{1 - \sqrt{\mathrm{KL}(P_+, P_-)/2}}{2} \ge \frac{1}{4}.$$

Hence

$$R_m^* \;\ge\; \frac{\sigma}{8\sqrt{m}},$$

so one valid explicit choice is $c_0 = 1/8$.

The second obstruction is local drift. Fix any $1 \le h \le m$ and define

$$\beta_h := \min\Big\{\zeta,\; \frac{\sigma}{4h^{3/2}}\Big\}, \qquad \mu_{t-j}^{\pm,h} := \pm\,\beta_h\,(h-j)_+, \qquad 0 \le j \le m-1,$$

where $(u)_+ := \max\{u, 0\}$. Each path belongs to the admissible class because its slope magnitude is at most $\beta_h \le \zeta$. The endpoint gap is

$$|\mu_t^{+,h} - \mu_t^{-,h}| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\mathrm{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2}\sum_{j=0}^{m-1}\big(2\beta_h(h-j)_+\big)^2 \le \frac{2\beta_h^2}{\sigma^2}\sum_{r=1}^{h} r^2 \le \frac{2\beta_h^2}{\sigma^2} h^3 \le \frac{1}{8}.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold $\beta_h h$, and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1-\mathrm{TV}(P_{+,h},P_{-,h})}{2} \ge \frac{1-\sqrt{\mathrm{KL}(P_{+,h},P_{-,h})/2}}{2} \ge \frac{3}{8}.$$

Therefore

$$R_m^* \;\ge\; \frac{3}{8}\beta_h h \;\ge\; \frac{3}{32}\min\bigl\{\zeta h,\; \sigma h^{-1/2}\bigr\}$$

which is again a fully explicit numerical lower bound. Since h was arbitrary,

$$R_m^* \;\ge\; \frac{3}{32}\sup_{1\le h\le m}\min\bigl\{\zeta h,\; \sigma h^{-1/2}\bigr\}\,.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume $1 \le h_\star := (\sigma/\zeta)^{2/3} \le m$. Choose an integer $\bar{h} \in [h_\star/2, h_\star]$. Then

$$\zeta\bar{h} \ge \frac{1}{2}\zeta h_\star = \frac{1}{2}\sigma^{2/3}\zeta^{1/3},$$

and

$$\sigma\bar{h}^{-1/2} \ge \sigma h_\star^{-1/2} = \sigma^{2/3}\zeta^{1/3}.$$

Therefore

$$\min\bigl\{\sigma\bar{h}^{-1/2},\; \zeta\bar{h}\bigr\} \ge \frac{1}{2}\sigma^{2/3}\zeta^{1/3},$$

and the ramp bound gives

$$R_m^* \ge \frac{3}{64}\sigma^{2/3}\zeta^{1/3}.$$

So one admissible explicit constant in the critical-window regime is $c = 3/64$.

Remark 2.9 (Scope of the lower bound). The lower bound is parametric only in the testing construction: it uses a Gaussian location subclass whose mean path is itself W_2-Lipschitz, and the distance in the lower bound is the endpoint W_2 error (Euclidean on that subclass). That is enough to make the result minimax for any larger class containing that subclass, including the paper's general W_2-Lipschitz path class. So the proposition is not claiming that every path is Gaussian; it is claiming that the worst case already appears inside the Gaussian mean-shift subproblem. Equivalently, since the Gaussian location family is a subfamily of the ambient W_2-Lipschitz class at the same drift rate ζ, any lower bound proven on that subclass is automatically a lower bound for the larger class.

Remark 2.10 (Measurement layer). Debiased Sinkhorn divergence serves as a practical measurement layer when the underlying geometry must be estimated from samples. That computational choice supports the tracking law, but the law itself is about the horizon of useful memory, not about any particular OT solver.

3 Empirical Validation

The experiments test a single claim: under continuous drift, memory must be regulated by temporal validity rather than by statistical evidence of abrupt change alone.

Empirically, the paper separates three layers that are often collapsed: horizon geometry, horizon tracking, and predictive payoff. The Gaussian sweep and lag–variance frontier validate geometry. The continuous-drift, temporal-validity, and alternating-timescales benchmarks probe tracking. The horizon cost curve assesses predictive payoff. A compact piecewise benchmark is retained only as a phase-wise calibration check.

In the experiments, UMR is the useful-memory regulator and ADWIN is the backend detector instantiated in the benchmarks.

3.1 Law Validation

We begin with the Gaussian sweep used to validate the finite-memory law. The setup produces the characteristic U-curve: for small windows, error is dominated by variance; for large windows, error is dominated by staleness. The empirical minimum follows the predicted cube-root scaling, and EWMA behaves as the natural finite-memory analogue rather than as a competing theory.

To make the comparison statistically explicit, we also ran a paired bootstrap over the 20 synthetic seeds (50,000 resamples). Table 1 reports the tail-MAE mean and 95% interval for each method, together with the paired difference relative to ADWIN+ UMR. The cube-cap remains better than ADWIN, EWMA, and the long fixed window on this benchmark, while the short fixed window remains the strongest baseline.

Table 1: Seed-wise bootstrap summary for the main synthetic benchmark. The first column block reports tail-MAE means with 95% bootstrap intervals across the 20 seeds; the second block reports paired differences versus ADWIN+ UMR.

Method	Tail MAE [95% CI]	Δ vs cube [95% CI]
Fixed-100	0.0849 [0.0815, 0.0884]	−0.0130 [−0.0144, −0.0115]
Fixed-500	0.2388 [0.2274, 0.2501]	+0.1409 [+0.1319, +0.1504]
EWMA	0.1292 [0.1263, 0.1319]	+0.0313 [+0.0278, +0.0350]
ADWIN	0.1757 [0.1662, 0.1853]	+0.0779 [+0.0704, +0.0857]
ADWIN + UMR	0.0978 [0.0942, 0.1016]	–

3.2 Lag–Variance Frontier

The U-curve becomes more informative when viewed as a frontier in horizon space. The next figure plots tail error against effective memory horizon for a fixed-window sweep and overlays the operating points of the main methods. The curve is asymmetric: on the left, extra memory still buys visible noise reduction; on the right, extra memory mainly adds age. The knee is where that trade stops paying off.

3.3 Over-Memory Under Continuous Drift

The next benchmark focuses on the operational failure mode. We compare five methods on a smooth drifting stream: a short fixed window, a long fixed window,

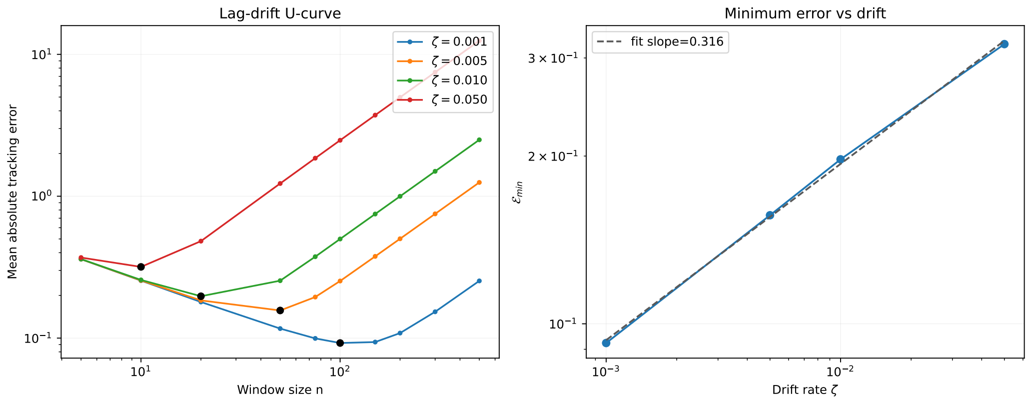

Figure 1: Cube-root validation on the Gaussian sweep. Left: tracking error as a function of window size, showing the predicted U-shape for multiple drift rates. Right: the minimum achievable error follows an approximate $\zeta^{1/3}$ law. This figure validates the structural trade-off between variance and staleness that underlies the rest of the paper.

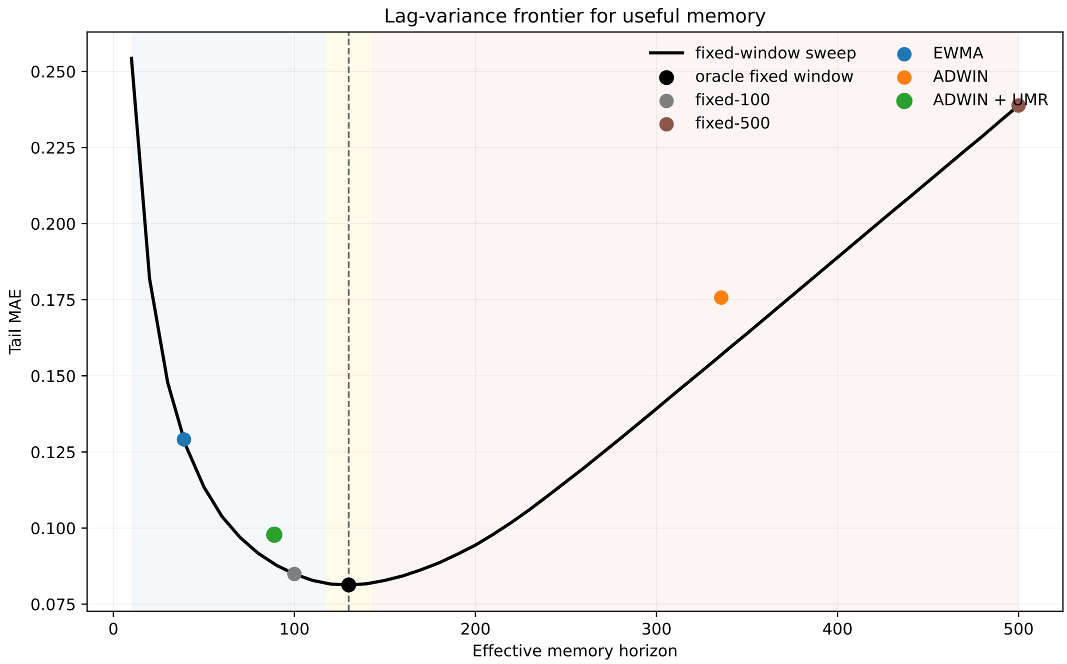

Figure 2: Lag–variance frontier for useful memory. The fixed-window sweep traces the empirical frontier: short horizons are noisy, long horizons are stale, and the knee marks the useful-memory optimum. The frontier is asymmetric because the gain from more memory saturates while the cost of age keeps accumulating. ADWIN sits far to the stale side, whereas ADWIN+ UMR lies close to the frontier knee.

EWMA, ADWIN, and the cube-root capped variant. The question is when memory becomes stale before changepoint evidence accumulates.

The key phenomenon is the separation between cap activation and the Hoeffding test. The cap can activate while the test remains silent, which makes memory validity distinct from the evidence threshold for abrupt change.

Table 2: Continuous-drift benchmark under a smooth drift rate. The key gap is between statistical evidence and temporal validity: ADWIN keeps a much wider window than the cube-root regulator, while the long fixed window collapses under lag.

Method	Tail MAE	Mean horizon	Drift events	Cap events
Fixed-100	0.0849	100.0	–	–
Fixed-500	0.2388	500.0	–	–
EWMA	0.1291	39.0	–	–
ADWIN	0.1757	335.8	3.9	0.0
ADWIN + UMR	0.0978	88.9	0.0	1374.7

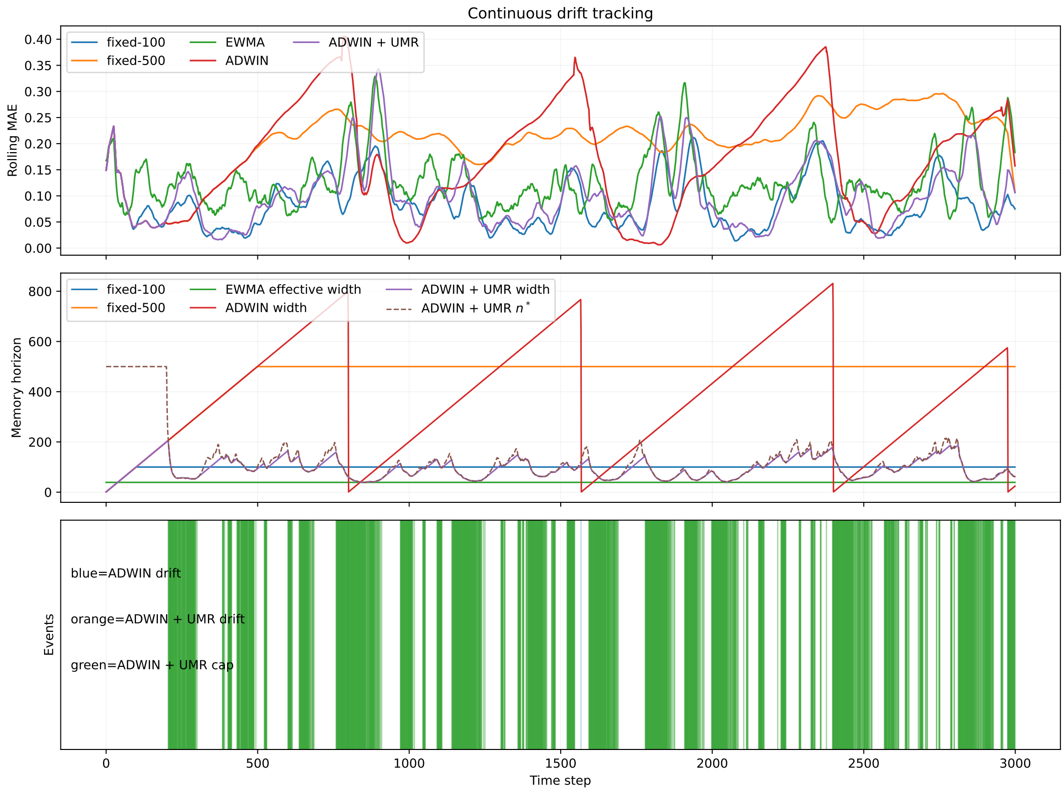

Figure 3: Continuous-drift benchmark. Fixed-500 is the stale-memory baseline; ADWIN retains far more memory than the oracle-scale horizon; ADWIN+UMR pulls the effective horizon toward the useful-memory scale and exposes a 'cap-only' regime, where memory becomes operationally stale before statistical changepoint evidence appears.

3.4 Temporal Validity Gap

The gap becomes more explicit when the drift rate varies. The next figure plots the first cap activation of ADWIN+UMR and the first changepoint decision of ADWIN as the drift rate changes. In the smooth-drift regime, the cap activates much earlier than the detector; as the drift becomes faster, the gap shrinks but does not disappear.

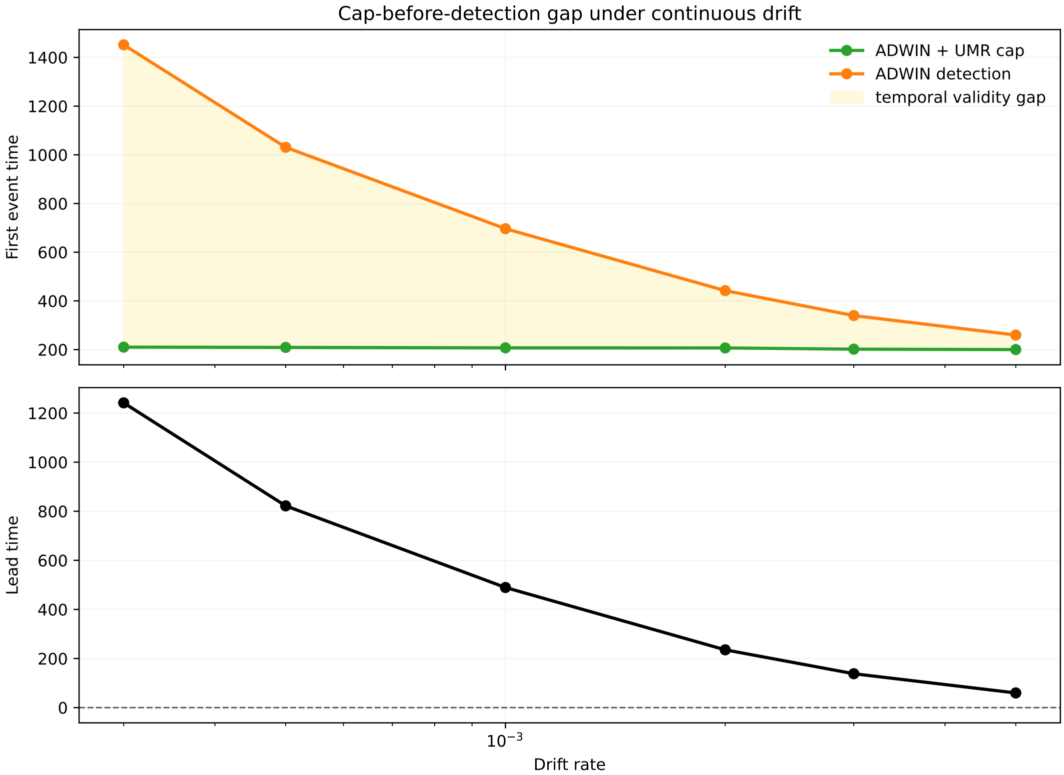

Figure 4: Temporal validity gap across drift rates. The top panel compares the first ADWIN+ UMR cap activation with the first ADWIN detection. The bottom panel shows the resulting lead time. Under slow drift, memory becomes operationally stale long before changepoint evidence is available; under faster drift, the gap narrows but remains positive.

3.5 Fixed-Horizon Instability and Recovery Asymmetry

The next benchmark makes fixed-horizon instability visible. The stream alternates between slow drift, fast bursts, recovery plateaus, and micro-bursts with irregular durations, so the local oracle horizon need not be globally stable. In that setting, a single fixed window can look reasonable on average while remaining locally misaligned for long stretches. This benchmark separates three questions that are often conflated: whether the oracle horizon is time-varying, whether an online regulator can track that changing scale, and how far each method remains from the local oracle horizon over time.

This benchmark makes two things visible. First, horizon validity can vary even when aggregate prediction error stays competitive. Second, the adjustment is asymmetric: contraction of the useful horizon is easier to trigger than re-expansion after a low-drift stretch. The pattern is operational evidence of path dependence in temporal memory adaptation. Proposition 3.1 formalizes the same asymmetry in the idealized EMA-to-horizon map.

This is the paper's secondary indexable contribution: a horizon-recovery asymmetry for EMA-driven regulators, distinct from the cube-root scaling law itself.

Proposition 3.1 (Asymmetric Horizon Recovery)**.** *Let the drift estimator satisfy*

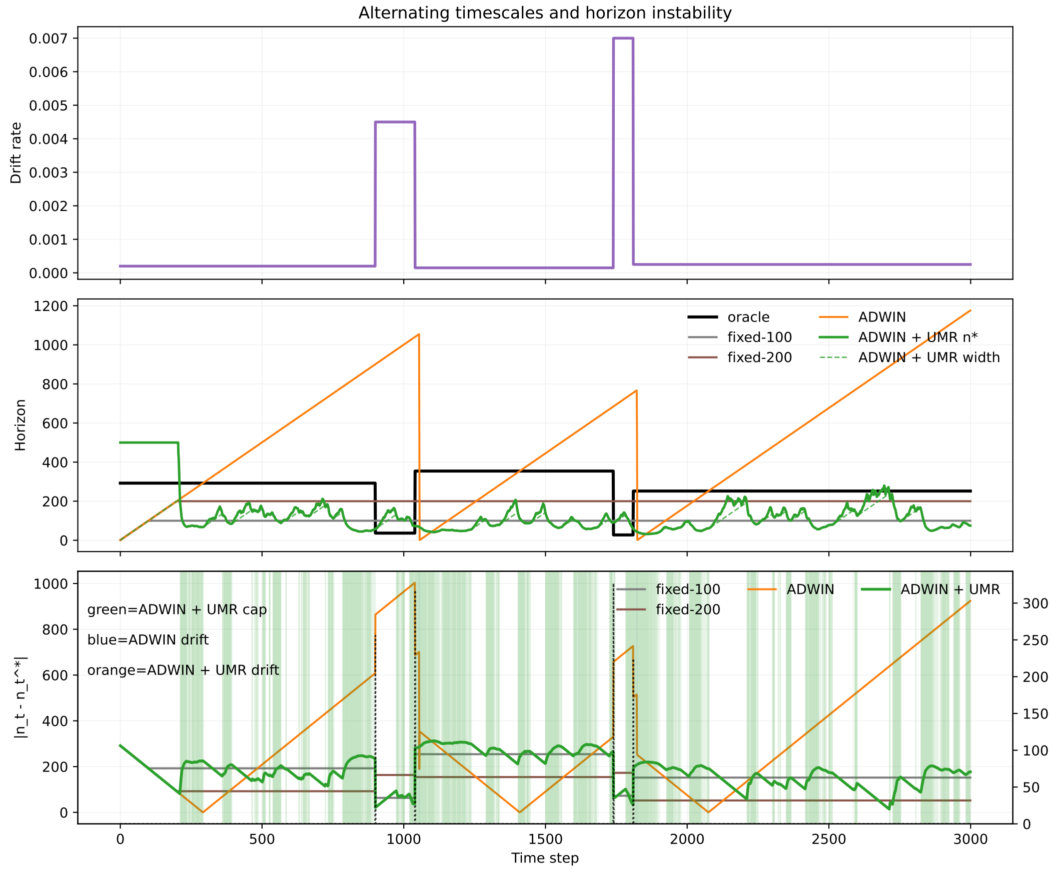

Figure 5: Alternating-timescales instability. The top panel shows the drift schedule, the middle panel compares the local oracle horizon with fixed windows and ADWIN+ UMR, and the bottom panel shows horizon regret over time. The figure exposes local horizon instability under regime changes.

the EMA recursion

$$\hat{\zeta}_t \;=\; \alpha\,\Delta_t + q\,\hat{\zeta}_{t-1}, \qquad q := 1-\alpha \in (0,1),$$

with $\Delta_t = \delta_{\mathrm{slow}}$ *in the slow regime and* $\Delta_t = M$ *in the burst regime. Assume the horizon map is unclipped on* $[\delta_{\mathrm{slow}}, M]$ *so that*

$$n^*(\delta) \;=\; \left(\frac{C_K}{\delta}\right)^{2/3}$$

throughout the transition. Write $n_b := n^*(M)$ *and* $n_s := n^*(\delta_{\mathrm{slow}})$*. For any* $0 < \varepsilon < n_s - n_b$*, define the inverse thresholds*

$$\delta_b(\varepsilon) := C_K\,(n_b+\varepsilon)^{-3/2}, \qquad \delta_s(\varepsilon) := C_K\,(n_s-\varepsilon)^{-3/2}.$$

Then the exact recovery time from the slow regime to the burst regime is

$$T_{\mathrm{contract}}(\varepsilon) \;=\; \left\lceil \frac{\log\big((M-\delta_{\mathrm{slow}})/(M-\delta_b(\varepsilon))\big)}{\log(1/q)} \right\rceil,$$

while, after a burst of duration T_{burst} steps, the exact recovery time back to the slow regime is

$$T_{\text{expand}}(\varepsilon) \;=\; \left\lceil \frac{\log\!\big((\hat{\zeta}_{\text{end}}-\delta_{\text{slow}})/(\delta_s(\varepsilon)-\delta_{\text{slow}})\big)}{\log(1/q)} \right\rceil,$$

where

$$\hat{\zeta}_{\text{end}} \;=\; M-(M-\delta_{\text{slow}})q^{T_{\text{burst}}}$$

is the burst-end drift estimate. If

$$T_{\text{burst}} \;\geq\; T_0(\varepsilon) \;:=\; \left\lceil \frac{\log(1-\rho_\varepsilon)}{\log q} \right\rceil, \qquad \rho_\varepsilon := \frac{\delta_s(\varepsilon)-\delta_{\text{slow}}}{M-\delta_b(\varepsilon)},$$

then $T_{\text{expand}}(\varepsilon) > T_{\text{contract}}(\varepsilon)$. Moreover, for small ε,

$$M-\delta_b(\varepsilon) = \tfrac{3}{2}C_K n_b^{-5/2}\,\varepsilon + O(\varepsilon^2), \qquad \delta_s(\varepsilon)-\delta_{\text{slow}} = \tfrac{3}{2}C_K n_s^{-5/2}\,\varepsilon + O(\varepsilon^2),$$

so the slow-side threshold distance is smaller by a factor $\big(\delta_{\text{slow}}/M\big)^{5/3}(1+o(1))$.

Proof. The EMA recursion has the closed form

$$\hat{\zeta}_{t_0+k} = \Delta^* + \big(\hat{\zeta}_{t_0}-\Delta^*\big)q^k$$

whenever the regime is constant with fixed point Δ^*. Hence after the transition from δ_{slow} to M,

$$\hat{\zeta}_{t_0+k} = M-(M-\delta_{\text{slow}})q^k,$$

and after the return to the slow regime,

$$\hat{\zeta}_{t_0+T_{\text{burst}}+k} = \delta_{\text{slow}} + (\hat{\zeta}_{\text{end}}-\delta_{\text{slow}})q^k.$$

Because $n^*(\delta)$ is monotone decreasing on the unclipped interval and its inverse is $g(n) = C_K n^{-3/2}$, the condition $|n^*(\hat{\zeta}_{t_0+k})-n_b| \leq \varepsilon$ is equivalent to $\hat{\zeta}_{t_0+k} \geq \delta_b(\varepsilon)$, which gives the displayed formula for $T_{\text{contract}}(\varepsilon)$. Likewise, $|n^*(\hat{\zeta}_{t_0+T_{\text{burst}}+k})-n_s| \leq \varepsilon$ is equivalent to $\hat{\zeta}_{t_0+T_{\text{burst}}+k} \leq \delta_s(\varepsilon)$, which yields $T_{\text{expand}}(\varepsilon)$.

The inequality $T_{\text{expand}}(\varepsilon) > T_{\text{contract}}(\varepsilon)$ holds whenever the burst lasts long enough that

$$1-q^{T_{\text{burst}}} \geq \rho_\varepsilon,$$

since this makes the numerator of T_{expand} larger than that of T_{contract} by the exact threshold relation above. The small-ε expansions follow by Taylor expansion of $n \mapsto C_K n^{-3/2}$ around n_b and n_s. □

Remark 3.2 (Effect of clipping). If clipping at $n_{\min}$ is active near the burst side, then contraction can only be faster. The observed asymmetry is therefore conservative: clipping weakens the theoretical barrier in the burst regime, but it cannot eliminate the longer slow- side recovery.

Table 3: Mean horizon regret over the first 50 steps after each regime change. The transition-window ablation shows that contraction and re-expansion are not symmetrical. Lower is better.

Method	Contraction	Expansion
fixed-100	63.3	254.2
fixed-200	163.3	154.2
ADWIN	888.8	435.8
ADWIN + UMR	58.4	307.7

3.6 Drift-EMA Sensitivity

The recovery asymmetry persists when the drift estimator itself is varied. The next sweep changes the EMA coefficient α in the drift estimator while keeping the alternating-timescales benchmark fixed. Larger α makes the drift estimate more reactive, reducing contraction regret while expansion regret stays persistently larger; the expansion-to-contraction ratio remains above one throughout the sweep and is higher at the larger α values. Appendix D then shows that this sensitivity can be reduced by an observable prefix-validation study for (d, α), which shifts the operating point of the regulator and helps interpret the sensitivity of the drift proxy.

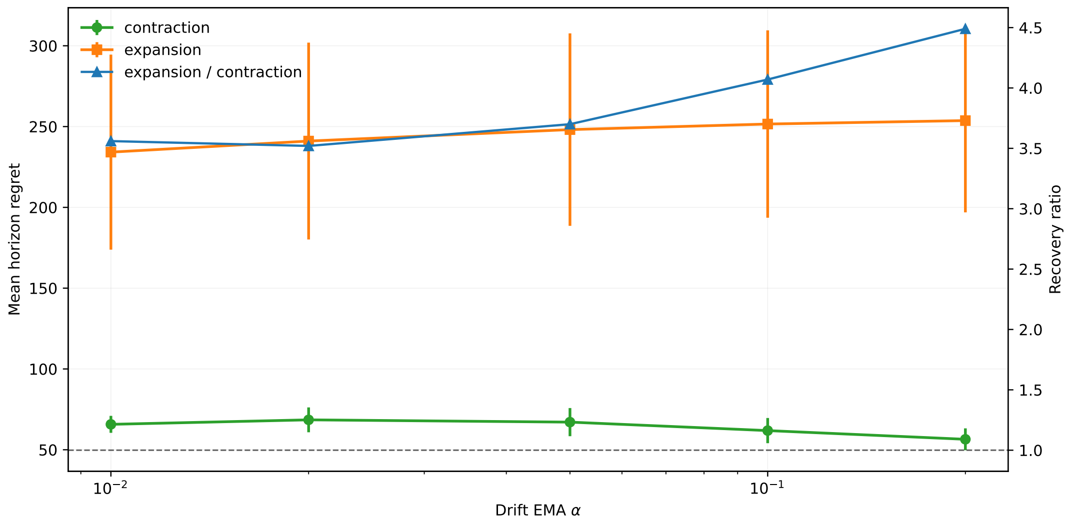

Figure 6: EMA sensitivity of horizon recovery on alternating timescales. The drift-EMA coefficient changes how quickly the regulator reacts to bursts, but the asymmetry remains: expansion regret stays above contraction regret across the sweep, and the ratio is larger at higher α in this benchmark.

3.7 Horizon Cost Curve

The next question is when horizon misalignment becomes operationally expensive. For each backend, we compare its instantaneous horizon n_t with the local cube-root target n_t^* and plot the excess error $E_t - E_t^\star$ against both the absolute gap $|n_t - n_t^*|$

and the relative gap $|\log(n_t/n_t^*)|$. Here $E_t^\star$ is the best error achieved at time t by the fixed-window oracle grid, and the curve is binned over all seeds and time points.

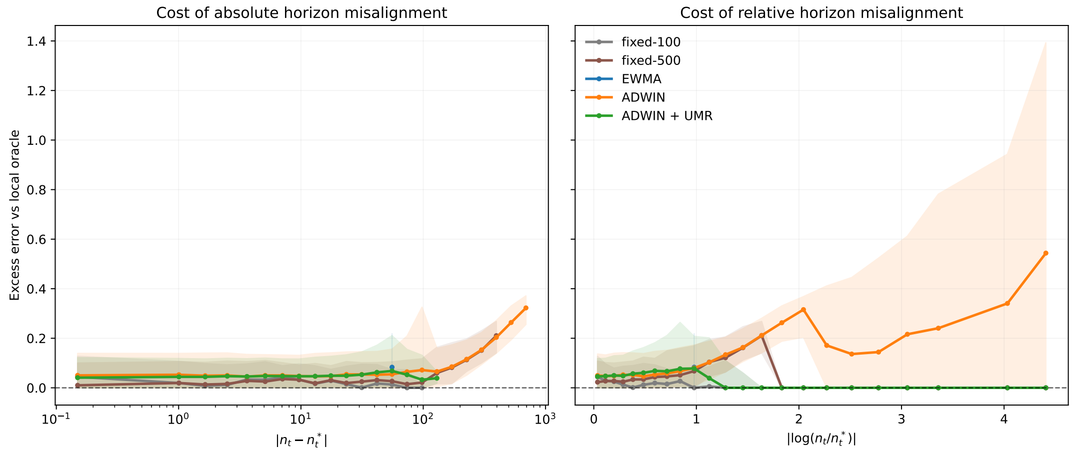

Figure 7: Excess error as a function of horizon misalignment. The relation is backend dependent rather than universal: fixed-100 stays inside a low-cost band, whereas fixed-500 and ADWIN pay substantially more once the gap widens. ADWIN+ UMR sits between these extremes. Some curves terminate early because the corresponding method has no observations beyond that gap range; the visible endpoints are data-limited, not clipped.

The main lesson is that the paper exposes a measurable tolerance band around the useful-memory scale, together with backend-specific degradation once that band is exceeded.

3.8 Phase-Wise Recovery Check

The piecewise-drift benchmark changes the local drift rate in three phases, and the adaptive horizon is compared with the offline oracle fixed window for each phase. The calibration is held fixed across phases; only the stream changes. The detailed calibration check is reported in appendix B. In the main text, the point is simply that one calibration remains in the right operational range when the drift regime changes discretely.

3.9 Representative Real-Stream Case Study

The synthetic experiments isolate the geometry of horizon control and carry the main claim of the paper. The real-data section plays a different role. Here we keep a single representative public stream, ELEC2, to check that the same temporal-validity story still appears once the drift process is no longer under experimental control. It is read as a case study, not as a ranking exercise. The full ELEC2/Bikes arena is provided in appendix E.

The real-data picture is consistent with the synthetic one, but the effect is mixed enough that it should not carry the main narrative. The point is simply that coarse horizon adaptation still matters on public data, while the payoff remains backend dependent.

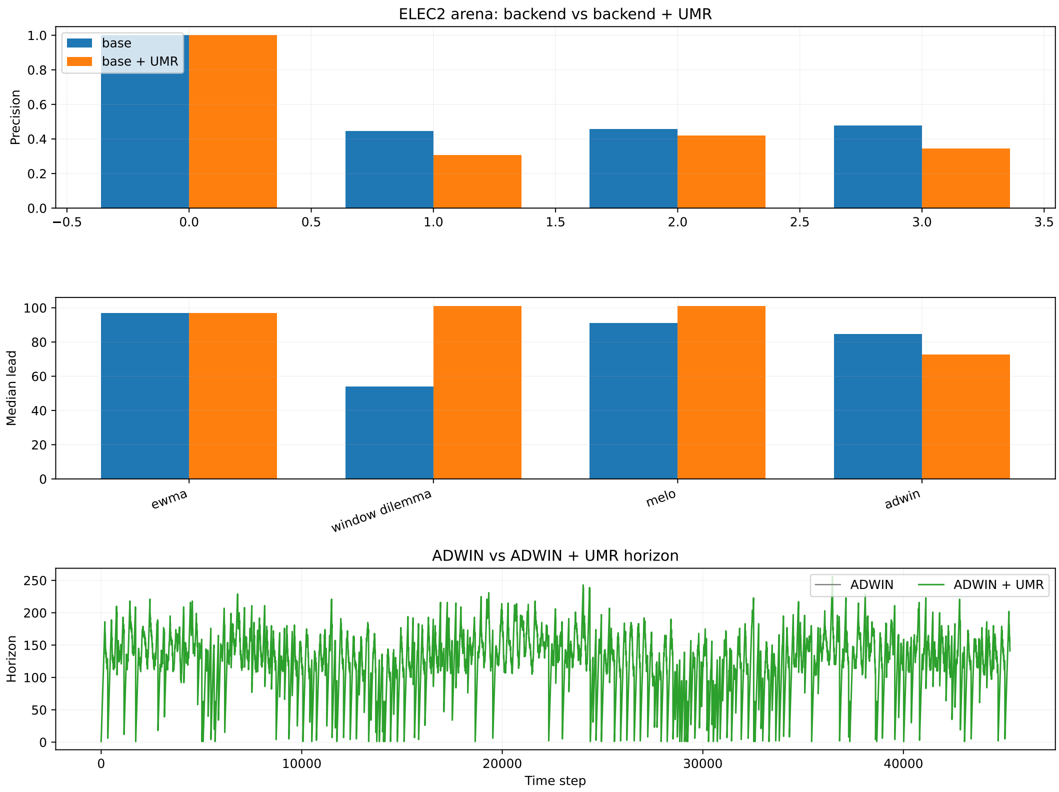

Figure 8: ELEC2 representative case study. The paired precision and median-lead comparison shows the same temporal-validity gap seen in the synthetic setting, but here the effect is read as a public-stream illustration rather than a leaderboard result.

3.10 Controlled Downstream Check

The downstream classification benchmark is reported in Appendix F. Its role is purely mechanistic: it checks whether the cap-only regime carries a measurable predictive cost when the backend is a windowed classifier rather than a drift detector. The effect is directionally consistent with the main claim but modest in magnitude, so we treat it as supporting evidence rather than central evidence.

3.11 What the Figures Show

Taken together, the figures support three claims. First, the cube-root law is a real geometric law for useful memory rather than a fitting artifact. Second, horizon tracking is distinct from changepoint detection: memory can become stale before statistical evidence is strong enough to declare a change. Third, horizon control has operational value, but its predictive payoff depends on the backend and on the drift regime. The ELEC2 case study reinforces the same temporal-validity gap on public data, but it is best read as a supporting check on the main synthetic result.

4 Related Work

Tracking under nonstationarity. Dynamic regret, online least-squares, and adaptive filtering all ask a closely related question: how much past evidence should remain relevant once the target is moving [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. The present paper keeps that question in distributional form and makes the memory horizon itself the main object of analysis.

Uncertainty-aware multi-horizon ensembles attack the same practical issue from a different angle: they train learners on several historical windows and then use calibrated uncertainty to weight the horizon-specific predictions [Rajput et al., 2025]. That framework is complementary to the present one. In our setting, the cube-root horizon can serve as a horizon prior or cap, but only within the same expert-mixing logic; the effect is benchmark-dependent and modest.

Remark 4.1 (Distinction from dynamic-regret analyses). The $\zeta^{1/3}$ exponent also appears in non-stationary stochastic optimization [Besbes et al., 2015, Cheung et al., 2019] and modern variance-dependent non-stationary bandits [Wang et al., 2024] and online tracking with path variation [Yuan and Lamperski, 2020]. Those results concern gradient-based methods on sequences of convex losses in Euclidean parameter space, where drift is measured by a variation budget on loss functions. The present law operates in a distinct setting: (i) drift is W_2-Lipschitz on the space of probability measures, not on a Euclidean loss gradient; (ii) the estimator class is causal averaging kernels, not parametric gradient methods; (iii) Proposition 2.8 establishes the $\zeta^{1/3}$ rate as a minimax lower bound for arbitrary window-restricted estimators, not only for averaging—an obstruction from finite memory itself, rather than from variation-budget constraints.

Jiang, Li, and Zhang analyze online stochastic optimization with a Wasserstein-based nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Our paper takes that geometry as background and specializes the question to finite-memory tracking, where the core object is the usable horizon rather than policy regret.

Taken together, this prior work makes the framing clear: the cube-root regime is not new as a scaling law in Wasserstein OCO, but here it becomes a structural limit of finite-memory tracking under drift, with practical consequences for window regulators and horizon-weighted ensembles.

Transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural because they respect the geometry of the signal space and behave well under transport of mass and support. Recent finite-sample results clarify how the constants depend on geometry and regularization, while leaving the exponent structure intact [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. Here, the geometry quantifies how much old evidence can be retained before it becomes stale.

Adaptive forgetting and horizon control. Classical smoothers such as EWMA, forgetting-factor RLS, and Kalman filtering all instantiate the same tension between variance reduction and responsiveness. What they often leave implicit is the temporal validity of the retained memory. The cube-root law makes that validity visible by turning the coherence-lag trade-off into an operational horizon selector.

The asymmetry result below is adjacent to this literature but not covered by it: EMA-based control papers typically analyze smoothing as a parameter-regularizing device, while changepoint methods such as ADWIN give detection guarantees rather than recovery-time guarantees. Our contribution is the gap between those two questions: once drift evidence is mapped into a horizon rule, re-expansion can be provably slower than contraction.

Classic variation-budget analyses in nonstationary OCO and bandits already exhibit the same cube-root lineage: optimal windows or step sizes scale with a two-thirds exponent, while the variation term enters with a one-thirds exponent in the resulting regret or tracking trade-off [Besbes et al., 2015, Cheung et al., 2019, Wang et al., 2024, Yuan and Lamperski, 2020]. The present paper should be read as a Wasserstein-distributional analogue of that line, not as a claim to invent the exponent structure itself. In that sense, it bridges statistical geometry (Wasserstein/Sinkhorn) and operational streaming policies: the geometry sets the measurement layer, while the horizon law sets the control layer.

This also complements the "Window Dilemma" view that concept-drift detection is ill-posed once the window scale itself is part of the problem [Hinder et al., 2023]: instead of treating windowing as a nuisance, we make temporal coherence explicit and use it to regulate the scale under continuous drift.

Keehan, Anderson, and Wiesemann [Keehan et al., 2025] study a closely related drift-aware weighting problem in Wasserstein distributionally robust optimization. Their result balances effective sample size against drift in the weighted empirical distribution. The present paper differs in object and goal: it asks when memory remains temporally valid, rather than how to optimize a robust weighting rule, and the relevant comparison with drift detection is the gap between when change becomes statistically detectable and when retained memory stops being operationally current.

The empirical section now includes a Window-Dilemma-style switcher and a MELO-style multi-horizon hedge as direct comparison baselines. Those additions do not replace the main claim of the paper, but they make the practical positioning sharper: the cube-root regulator is being compared not only to fixed windows and detectors, but also to explicit horizon-selection and expert-mixing alternatives.

Appendix E further adds a Rajput-style uncertainty- weighted horizon ensemble analogue on the public streams. There, UMR functions as a horizon cap rather than a new weighting rule, and the observed effect remains small and benchmark-dependent.

This makes the design choice explicit: single-horizon regulation is the natural fit when one believes there is a single locally meaningful memory scale and the main issue is keeping that scale temporally valid, whereas multi-horizon ensembles are more appropriate when different windows carry complementary uncertainty or the stream alternates between regimes. In our public-stream analogue, the gains from adding UMR to the ensemble are modest, so the comparison reads as complementary rather than substitutive.

Freshness and the Age of Information. The Age-of-Information literature asks how old the last received update is and how update policies should balance communication cost against information staleness [Kaul et al., 2012, Yates et al., 2021]. That question is posed for status signals transmitted over communication channels. The present paper answers the distributional analogue: when a target law evolves under W_2 drift, useful memory has a finite horizon and that horizon obeys the cube-root law. The connection is interpretive rather than formal, but it suggests a natural extension: policies for refreshing a distributional estimate, not only a scalar status signal.

5 Limitations and Scope

Scope of the law. The cube-root law is exact for the finite-memory averaging class analyzed in this paper, and the lower bound is stated for window-restricted estimators on a W_2-Lipschitz drift class. The lower bound itself is proved through a Gaussian location subclass and endpoint W_2 error, so its structural claim is worst-case and subclass-based rather than a universal theorem over all possible online estimators.

Dependence on local drift estimation. The operational regulator depends on a local estimate of drift. The current benchmark suggests that the remaining gap to the oracle comes mostly from drift estimation. At the same time, this is not just a weakness: the paper now shows that a short observable calibration prefix can materially improve the regulator, so drift estimation is part of the method rather than a hidden nuisance parameter. Better local geometry estimators may therefore sharpen horizon recovery without changing the law.

Geometry beyond the current metric. The present formulation uses Wasserstein geometry and a debiased Sinkhorn measurement layer when samples must be compared computationally. Other geometries may admit analogous trade-offs, but the constants and the calibration of useful memory may change.

What the experiments show. The current experiments are not a leaderboard over drift detectors. The public stream result is a representative case study, not a ranking exercise. It shows a specific operational distinction: a tracker can still have statistically insufficient changepoint evidence while already being stale in the sense of useful memory.

Scope of the comparison suite. The benchmark suite now includes a Window-Dilemma-style switcher and a MELO-style multi-horizon hedge, but it is still not a full survey of adaptive tracking methods. The present paper remains centered on the cube-root law and on the temporal-validity interpretation of memory, rather than on an exhaustive aggregation benchmark zoo.

On the current ELEC2 and Bikes arenas, adding UMR to the multi-horizon hedge, including the Rajput-style uncertainty-weighted analogue in Appendix E, changes results only modestly and in a benchmark-dependent way; it is therefore best read as a horizon prior, not as a universal replacement for uncertainty weighting.

6 Conclusion

How long can memory remain useful when the target distribution keeps moving? Under the drift class studied here, useful memory has a finite temporal horizon, and that horizon obeys a cube-root law. The governing object is a coherence-lag trade-off: preserving temporally coherent evidence requires limiting the lag induced by old memory.

The theoretical contribution is a decomposition of tracking error into a variance term and a staleness term, together with a lower bound showing that the resulting finite-memory floor is structural. The operational contribution is a memory regulator: a cube-root cap keeps the effective horizon near the oracle scale across drift regimes, while classical adaptive windows can still accumulate stale memory.

The strongest empirical lesson is that horizon geometry, horizon tracking, and predictive payoff are distinct objects. Horizon misalignment carries a nonlinear operational cost, but the transfer from gap to error depends on the backend and on the drift regime.

Empirically, the signature regime is 'cap-only': the memory horizon becomes operationally stale before there is enough statistical evidence for a changepoint alarm. That separation between evidence of change and temporal validity of memory is the main conceptual payoff of the paper. The alternating-timescales benchmark adds a second operational lesson: horizon contraction is typically easier to induce than horizon re-expansion, revealing an asymmetric form of memory hysteresis. That asymmetry is a secondary formal contribution: a recoverability gap in EMA-driven horizon regulators, separate from the cube-root law itself. A third operational lesson is implementation-level: the regulator is sensitive to the drift proxy, but that sensitivity is calibratable from observable prefix data rather than only by manual tuning.

The real-stream evidence is intentionally lighter: a representative ELEC2 case study in the main text, while the full ELEC2/Bikes arena is detailed in the appendix. It serves as a supporting check that the temporal-validity gap is not synthetic-only, but it is not the main evidential load of the paper.

Finite-memory systems are best judged by how long their retained evidence remains current enough to matter. The design consequence is clear: regulate useful memory and track the age at which evidence stops being current.

Final claim: useful memory under drift has geometry, but its predictive value is policy-dependent.

Code and reproducibility artifacts are available in the project repository [**?**].

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Backend + UMR Protocol

This appendix gives the concrete protocol used by the paper whenever a backend is paired with the useful-memory regulator (UMR). The code paths are: `drift/umr.py` for the drift proxy and horizon rule, `experiments/core/horizon_baselines.py` for horizon-capped estimators, `experiments/core/detectors.py` for the ADWIN+ UMR reset logic, and `experiments/core/umr_arena.py` for the shared arena composition.

A.1 Shared Regulator

At each time step, the regulator maintains a local drift estimate and a target horizon:

$$\Delta_t = \frac{|\bar{x}_t^{(d)} - \bar{x}_{t-d}^{(d)}|}{d}, \qquad \hat{\zeta}_t = \alpha \Delta_t + (1-\alpha)\hat{\zeta}_{t-1},$$

$$n_t^* = \operatorname{clip}\left(\left(\frac{C_K}{\max(\hat{\zeta}_t, 10^{-9})}\right)^{2/3}, n_{\min}, n_{\max}\right).$$

There is no explicit cooldown parameter. The only stabilization is clipping through $n_{\min}$ and $n_{\max}$, plus the EMA smoothing in $\hat{\zeta}_t$.

A.2 Horizon-Capped Backends

For estimators whose natural state is a window size or effective horizon (fixed-window mean, EWMA, Window-Dilemma-style switching, MELO-style expert mixing), the runtime horizon is simply clipped by the regulator:

$$h_t^{\text{eff}} = \min(h_t^{\text{backend}}, n_t^*).$$

The backend then computes its estimate using h_t^{eff} exactly as it would have used h_t^{backend}.

A.3 ADWIN + UMR Event Logic

For the detector wrapper, the backend is ADWIN on a scalar signal. The regulated version maintains a rolling buffer of the last $n_{\max}$ observations and applies the following logic:

1. Update the drift proxy and compute n_t^*.
2. Feed the new observation to ADWIN.
3. Read the current ADWIN width w_t.
4. If $w_t > n_t^*$, record a cap event, reset ADWIN, and re-feed the most recent n_t^* buffered observations to the fresh detector.
5. If the detector then reports drift, record a warning.

The cap event is therefore a horizon event, not a detector event. The detector warning is still ADWIN's own decision after the reset path.

A.4 Hyperparameters

The paper uses three parameter families.

Table 4: Synthetic benchmark defaults used by `run_cuberoot_adwin.py`.

Scenario	Default hyperparameters
Constant	`drift` = 0.001, `fixed_window` = 100, `fixed_long_window` = 500, `ewma_alpha` = 0.05, `adwin_delta` = 0.002, `Ck` = 1.0, `drift_window` = 100
Piecewise	`piecewise_drifts` = $(0.0005, 0.003, 0.001)$ with the same window, EWMA, ADWIN, and C_K settings as above
Alternating	`piecewise_drifts` = $(0.0002, 0.0045, 0.00015, 0.007, 0.00025)$, `piecewise_lengths` = $(900, 140, 700, 70, 1190)$, `fixed_long_window` = 200, and the same remaining defaults
Common	$n_{\min} = 10$, $n_{\max} = 500$

Synthetic core suite.

Real-stream arenas. The ELEC2 and Bikes arenas use the shared arena builder in `experiments/core/umr_arena.py` with:

- `fixed_window` = 100, `fixed_short_window` = 50, `fixed_long_window` = 300;
- `ewma_alpha` = 0.05;
- `block_size` = 24, `ema_alpha` = 0.03;
- `min_window` = 30, `max_window` = 300, `scale` = 1.25;
- `baseline_window` = 100, `prefix_length` = 2000;
- `adwin_delta` = 0.03;
- Page-Hinkley event detector: $\delta = 0.01$, threshold 84.0, $\alpha = 0.9999$;
- Window-Dilemma windows $(50, 100, 300)$ with quantiles $(0.33, 0.67)$;
- MELO-style expert windows $(30, 50, 100, 200, 300)$ with learning rate 6.0 and discount 0.995.

Drift-proxy calibration. When C_K is not fixed externally, the code estimates it from a stationary prefix via `calibrate_umr_constant`, i.e. standard deviation times the square root of prefix length. The drift-proxy appendix additionally reports the prefix-validated (d, α) sensitivity study, but that is a robustness study rather than a replacement for the fixed-regulator core.

A.5 Reproducibility Pseudocode

1. **Input:** stream $x_{1:T}$, backend B, prefix length m, block size d, EMA coefficient α, bounds $[n_{\min}, n_{\max}]$, constant C_K.
2. For each $t = 1, \ldots, T$:
   (a) Update Δ_t from the last two blocks of length d.
   (b) Update $\hat{\zeta}_t = \alpha\Delta_t + (1-\alpha)\hat{\zeta}_{t-1}$.
   (c) Compute $n_t^* = \mathrm{clip}((C_K/\max(\hat{\zeta}_t, 10^{-9}))^{2/3}, n_{\min}, n_{\max})$.
   (d) If B is horizon-based, set the effective horizon to $\min(h_t^B, n_t^*)$ and evaluate the backend with that horizon.
   (e) If B is ADWIN, update the detector; if its width exceeds n_t^*, reset it and re-feed the last n_t^* buffered observations; then record any ADWIN warning.
3. **Output:** backend estimate, warnings, n_t^*, drift proxy, and cap events.

The exact implementation details are in the source files listed at the top of this appendix.

B Phase-Wise Recovery Details

This appendix presents the phase-wise calibration check for the UMR-regulated ADWIN benchmark. The result is that the same calibration stays in the right operational range when the drift regime changes discretely.

Table 5: Oracle horizon recovery on piecewise drift. The same calibrated cap tracks the oracle scale across regimes: it is close in the fast-drift phase and remains within the right order of magnitude in the smoother phases.

Drift	Oracle window	Regulated n^*	Oracle MAE	ADWIN + UMR MAE
0.0005	200	191.2	0.0741	0.0874
0.003	50	58.1	0.1258	0.1381
0.001	100	89.7	0.0876	0.1033

C Finite-Sample Geometry Check

This appendix records a numerical sanity check for the finite-sample term in lemma 2.2. We estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a log–log slope in n. We also check the Sinkhorn proxy in one dimension at a few fixed values of ε.

The point of this check is not a sharp asymptotic theorem. It is to confirm the paper's modeling choice: the low-dimensional raw-Wasserstein regime is compatible with an $n^{-1/2}$ term, while the Sinkhorn proxy behaves as a fixed-ε measurement layer with a calibration constant that depends on geometry and on regularization.

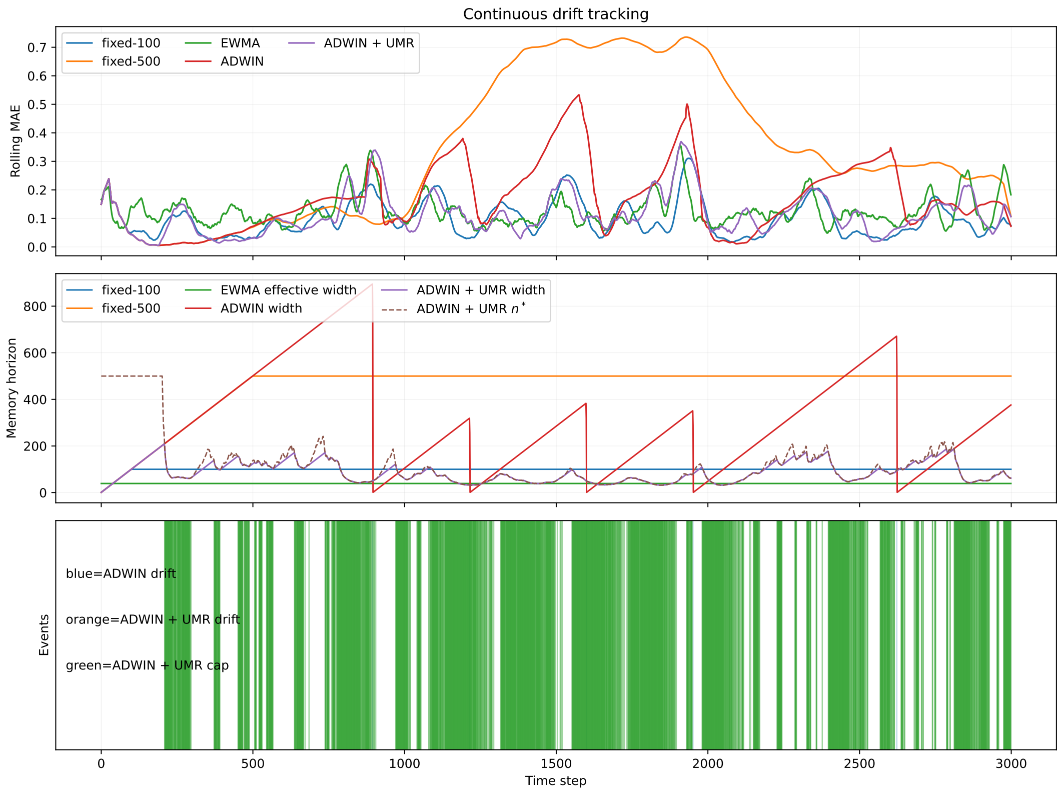

Figure 9: Piecewise-drift recovery. The same regulator calibration follows the phase-dependent oracle window closely enough to preserve the right operational scale in each regime.

Table 6: Empirical scaling check for the finite-sample term. The bounded-support W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, and the slope degrades as intrinsic dimension increases. The Sinkhorn proxy is stable for fixed ε, with the prefactor changing as ε varies.

Setting	Estimated slope	Comment
W_2, $d = 1$	-0.48	close to $-1/2$
W_2, $d = 2$	-0.44	slower decay
W_2, $d = 4$	-0.28	visibly slower
W_2, $d = 5$	-0.22	visibly slower
Sinkhorn $\varepsilon = 0.50$	-1.34	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-1.09	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.88	proxy constant changes

C.1 Sinkhorn ε Ablation and Implied n^* Inflation

The same measurement layer also clarifies how ε and sample size affect the cube-root calibration constant. In the Gaussian shift experiment used for the Sinkhorn runtime figure, the debiased Sinkhorn estimator overestimates the true $W_2^2 = 0.25$ by an amount that is strongly dimension dependent and decreases with sample size only when ε is not too small.

In the high-dimensional $d = 256$ slice, the empirical bias is nearly flat in n for

$\varepsilon \in \{0.02, 0.05, 0.2\}$, whereas at $\varepsilon = 1.0$ it decays noticeably with n (roughly from 0.479 at $n = 25$ to 0.145 at $n = 100$). On the same slice, the implied cube-root horizon inflation is about 1.36× to 4.24× across the tested grid, with the smallest inflation occurring at large ε and larger sample size.

If the signed bias is propagated linearly into the horizon rule, then the induced calibration error scales as $n^*_{\text{hat}}/n^* \approx (1+b)^{2/3}$ when the proxy overestimates the underlying W_2^2 by a factor $(1+b)$. Under this approximation, the tested $d = 256$ slice gives an n^* inflation between about 1.36× and 4.24× over the sampled grid.

Table 7: Sinkhorn ε ablation on the Gaussian shift calibration experiment. Reported values are signed biases relative to the true $W_2^2 = 0.25$ and the corresponding multiplicative inflation of the cube-root horizon under the linearized approximation $n^*_{\text{hat}}/n^* \approx (1+b)^{2/3}$.

Setting	Signed bias	Estimated W_2^2	Implied n^* factor
$d = 256$, $n = 25$, $\varepsilon = 1.0$	+0.460	0.710	2.01×
$d = 256$, $n = 100$, $\varepsilon = 1.0$	+0.139	0.389	1.34×
$d = 256$, $n = 200$, $\varepsilon = 1.0$	+0.073	0.323	1.19×
$d = 256$, $n = 25$, $\varepsilon = 0.05$	+1.797	2.047	4.06×
$d = 256$, $n = 100$, $\varepsilon = 0.05$	+1.765	2.015	4.02×
$d = 256$, $n = 200$, $\varepsilon = 0.05$	+1.748	1.998	4.00×

D Online Drift Estimation and Sensitivity

The dynamic horizon regulator uses a short-window drift proxy built from block differences. For block length d, let $\bar{x}_t^{(d)}$ denote the mean of the most recent block and $\bar{x}_{t-d}^{(d)}$ the mean of the previous block. The instantaneous drift proxy is

$$\Delta_t = \frac{\left|\bar{x}_t^{(d)} - \bar{x}_{t-d}^{(d)}\right|}{d},$$

and the smoothed estimate follows the EMA recursion

$$\widehat{\zeta}_t = \alpha\,\Delta_t + (1-\alpha)\,\widehat{\zeta}_{t-1}.$$

The regulator then sets

$$n_t^* = \operatorname{clip}\left(\left(\frac{C_K}{\max(\widehat{\zeta}_t, 10^{-9})}\right)^{2/3}, n_{\min}, n_{\max}\right).$$

Table 8 reports the sensitivity sweep already used in the paper. Larger α makes the horizon react faster to bursts, which lowers contraction regret but leaves expansion regret larger; the expansion-to- contraction ratio stays above one throughout.

The appendix complements the main-text proposition by giving the concrete estimator used in the benchmarks and the empirical sweep used to check that the asymmetry persists as the drift filter is retuned.

Table 8: Drift-EMA sensitivity on the alternating-timescales benchmark. The ratio remains above one across the sweep, so re-expansion is slower than contraction for every tested α.

α	Contraction	Expansion	Ratio	Tail MAE
0.01	65.73	234.18	3.56	0.0938
0.02	68.51	241.03	3.52	0.0923
0.05	67.10	248.12	3.70	0.0925
0.10	61.86	251.57	4.07	0.0934
0.20	56.48	253.70	4.49	0.0940

D.1 Robustness of the EMA Drift Proxy

Let the block-difference proxy satisfy $\Delta_t = \zeta_t + \xi_t$, where ξ_t is a zero-mean noise term with $\mathrm{Var}(\xi_t) \le \sigma_\Delta^2$ and the EMA recursion is

$$\widehat{\zeta}_t = \alpha\Delta_t + (1-\alpha)\widehat{\zeta}_{t-1}.$$

If $\zeta_t \equiv \zeta$ is constant, then

$$\mathbb{E}[\widehat{\zeta}_t] = \zeta + (1-\alpha)^t(\widehat{\zeta}_0 - \zeta), \qquad \mathrm{Var}(\widehat{\zeta}_t) \le \frac{\alpha}{2-\alpha}\,\sigma_\Delta^2.$$

So smaller α reduces variance but increases lag, while larger α reacts faster but amplifies noise.

Since the regulator uses

$$n_t^* = \left(\frac{C_K}{\widehat{\zeta}_t}\right)^{2/3},$$

the relative horizon error is first-order sensitive to relative drift-estimation error:

$$\frac{\delta n_t^*}{n^*} \approx -\frac{2}{3}\frac{\delta\widehat{\zeta}_t}{\zeta}.$$

Thus persistent overestimation of ζ induces under-memory, and persistent underestimation induces over-memory. If the proxy has multiplicative bias $\mathbb{E}[\Delta_t] = (1+b)\zeta$ with $b > 0$, then asymptotically

$$\frac{\widehat{n}^*}{n^*} = (1+b)^{-2/3},$$

so a 10% overestimate of drift shrinks the target horizon by about 6.5%.

For persistent overestimation, the same calculation gives systematic under-memory. For example, with a 10% positive bias the asymptotic horizon ratio is $1.1^{-2/3} \approx 0.94$, and with a 25% bias it is about 0.86.

D.2 Prefix-Validated Auto-Calibration

The operational fix is to choose (d, α) on a short calibration prefix by minimizing an *observable* prequential score over a small grid. For each candidate pair (d, α), run

Table 9: EMA robustness check for a constant drift proxy with additive noise. The empirical mean matches the target drift, the empirical variance matches the closed-form $\alpha/(2-\alpha)$ law, and the implied horizon distortion stays close to one when the proxy is unbiased.

α	$\mathbb{E}[\widehat{\zeta}_t]$	$\mathrm{Var}(\widehat{\zeta}_t)$	Theory	$\mathbb{E}[\widehat{n}^*]/n^*$
0.01	1.0000	0.00031	0.00031	1.0002
0.02	0.9999	0.00063	0.00063	1.0004
0.05	1.0000	0.00160	0.00160	1.0009
0.10	1.0000	0.00329	0.00329	1.0018
0.20	1.0002	0.00694	0.00694	1.0038

the regulated estimator on the prefix, generate the one-step-ahead forecast $\hat{x}^{(d,\alpha)}_{t|t-1}$, and score it by

$$S(d,\alpha) = \frac{1}{T_0-1}\sum_{t=1}^{T_0-1}\big(x_{t+1} - \hat{x}^{(d,\alpha)}_{t|t-1}\big)^2.$$

The selected pair minimizes this prequential MSE on the prefix and is then used for the full run. This is implementable online because it only uses observed data and the regulator's own forecasts.

Table 10: Prefix-validated auto-calibration on the synthetic benchmark. The calibration prefix chooses (d,α) by observable one-step-ahead prequential MSE, then the selected parameters are run on the full stream. This is a sensitivity study: the selected pair shifts the operating point, and on this calibration set it improves tail MAE relative to the fixed default.

Scenario	Mean α	Mean d	Tail MAE (default)	Tail MAE (calibrated)
Constant drift	0.19	50.0	0.2087	0.1224
Alternating drift	0.11	62.5	0.1020	0.0890

The failure mode is also visible: if the proxy is overreactive, the selected window contracts too aggressively and under-memory appears in the tail; if it is too inertial, the horizon lags the drift and over-memory follows. The calibration rule reduces that sensitivity by choosing the grid point that best predicts the next-step observations on the prefix.

The implemented self-calibrating rule is therefore simple: keep a small grid of block scales and EMA coefficients, score each candidate by prequential one-step MSE on a short prefix, then deploy the best candidate for the remainder of the stream. For non-Lipschitz local behavior, the grid should include at least one short and one longer block scale so that bursty changes do not force a single timescale to carry the entire burden. This is an operational refinement, not a replacement for the fixed-regulator results in the main text.

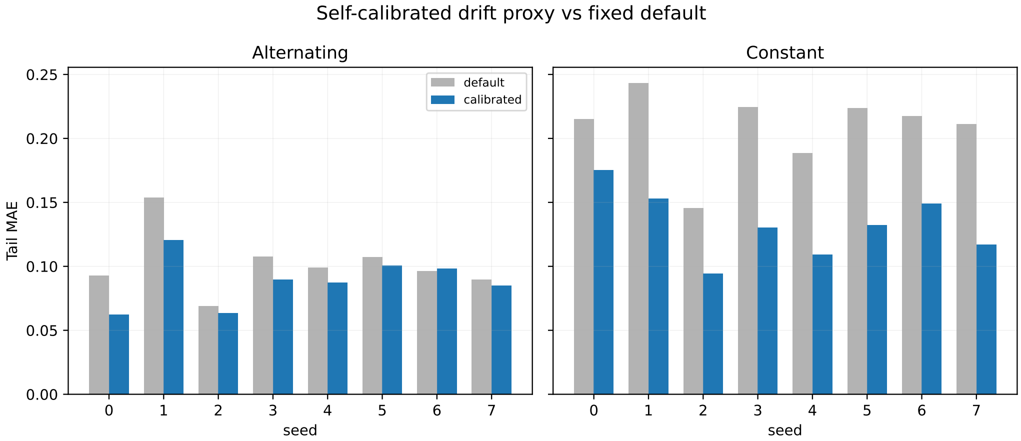

Figure 10: Prefix-validated self-calibration of the drift proxy. The selected (d, α) shifts the operating point of the regulator; on this calibration study it improves tail MAE relative to the fixed default.

E Full Real-Stream Arena

This appendix collects the complete public-stream robustness check summarized in the main text. ELEC2 appears there as the representative case study; here we add the Bikes counterpart and the paired summary table for both datasets.

Table 11: Full real-stream robustness summary. Precision is matched-warning precision against Page-Hinkley events; median lead is the median warning lead in steps. Each row compares a backend with and without UMR.

Dataset	Method	Precision	Median lead	Leads
ELEC2	EWMA	1.000	97.0	1
ELEC2	EWMA + UMR	1.000	97.0	1
ELEC2	Window Dilemma	0.446	54.0	29
ELEC2	Window Dilemma + UMR	0.305	101.0	29
ELEC2	MELO-style	0.458	91.0	11
ELEC2	MELO-style + UMR	0.419	101.0	18
ELEC2	ADWIN	0.478	84.5	54
ELEC2	ADWIN + UMR	0.345	72.5	10
Bikes	EWMA	0.146	132.0	59
Bikes	EWMA + UMR	0.146	132.0	59
Bikes	Window Dilemma	0.127	108.5	76
Bikes	Window Dilemma + UMR	0.128	125.0	76
Bikes	MELO-style	0.141	147.5	76
Bikes	MELO-style + UMR	0.129	147.0	71
Bikes	ADWIN	0.526	92.0	251
Bikes	ADWIN + UMR	0.520	82.0	141

The table shows the same mixed story across both streams: UMR can improve the temporal-validity side of the comparison, but the precision effect is backend-

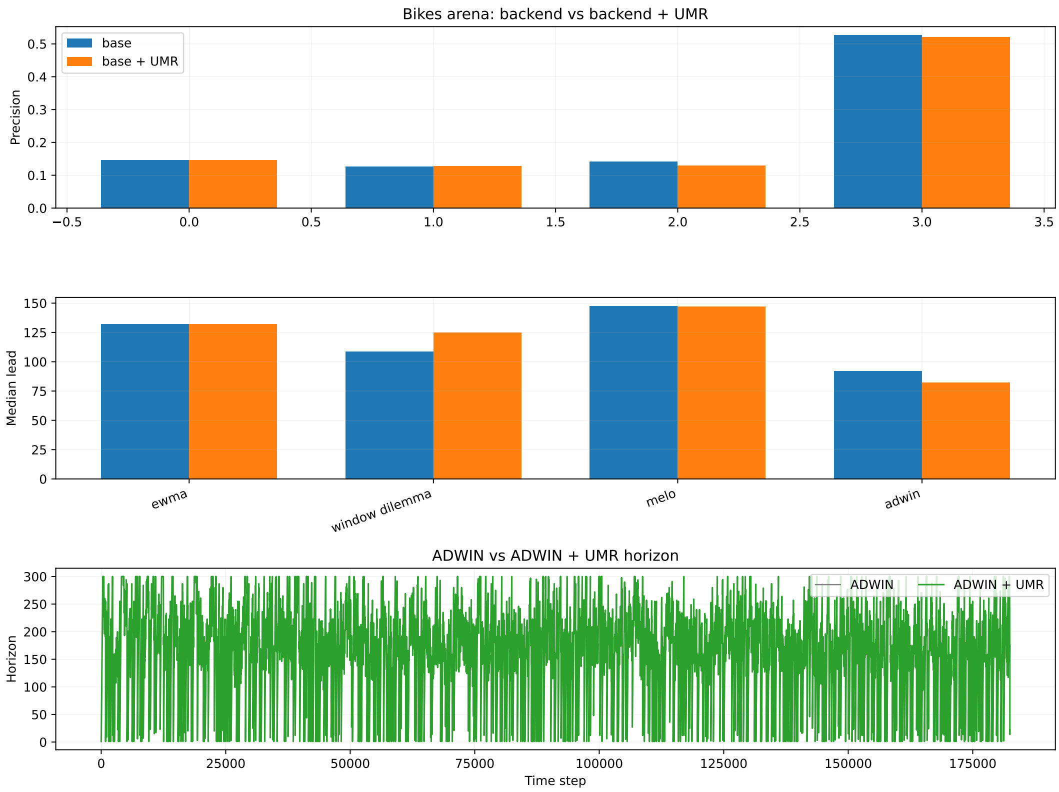

Figure 11: Bikes counterpart in the full real-stream arena. The same paired precision and median-lead comparison appears here, but the result is read as a robustness check rather than a benchmark claim.

dependent and sometimes negative. That is why the full arena is supporting evidence rather than the core argument.

E.1 Rajput-style uncertainty-weighted horizon ensemble

We also implemented a faithful algorithmic analogue of the Rajput et al. setup using fixed random-feature Bayesian regressors trained on rolling windows of historical context. The ensemble combines horizon-specific learners with inverse variance weighting, and the UMR regulator enters as a horizon cap over the active buffers. This is not the exact DIII-D/DGPA stack from the paper, but it is the closest compatible benchmark available in this repository.

The takeaway is consistent with the main text: UMR is compatible with a multi-horizon uncertainty-weighted ensemble, but in this public-stream analogue it acts as a weak horizon prior rather than a dominant improvement. On ELEC2, the single best horizon remains best; on Bikes, uncertainty weighting helps slightly, and UMR changes the outcome only marginally.

F Controlled Downstream Check

This appendix records the controlled synthetic downstream benchmark used to probe the predictive cost of temporal invalidity. The setup is intentionally minimal: a

Table 12: Rajput-style public-stream analogue. MAE is reported on the held-out test segment after prequential calibration. UMR acts as a horizon cap inside the ensemble. The effect is small and benchmark-dependent.

Dataset	Method	MAE	Relative change vs single
ELEC2	Single horizon	0.7565	0.0%
ELEC2	Naive ensemble	0.7681	+1.5%
ELEC2	UQ ensemble	0.7662	+1.3%
ELEC2	UQ ensemble + UMR	0.7648	+1.1%
Bikes	Single horizon	0.8212	0.0%
Bikes	Naive ensemble	0.8109	-1.3%
Bikes	UQ ensemble	0.8091	-1.5%
Bikes	UQ ensemble + UMR	0.8092	-1.5%

phasewise drifting binary classification stream, a windowed logistic backend, and four policies comparing long memory, short memory, ADWIN-driven memory, and UMR-regulated memory. The comparison is restricted to the phase-2 cap-only interval, where UMR has already contracted but ADWIN remains silent.

Missing figure artifact

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Figure 12: Controlled downstream benchmark. Top: true drift rate. Middle: effective horizons and the cap-only interval. Bottom: rolling prequential accuracy. The representative seed shows a 3.12 point gain for UMR over Fixed-400 within the cap-only window.

Table 13: Controlled downstream benchmark summary. The reported delta is UMR minus Fixed-400 within the cap-only interval, averaged across six seeds.

Method	Global acc.	Cap-only acc.	Δ cap-only	Mean cap-only width
Fixed-400	0.9304	0.9040	0.00	293.3
Fixed-100	0.9598	0.9576	+5.36	293.3
ADWIN	0.9340	0.9040	0.00	293.3
UMR	0.9415	0.9205	+1.66	293.3

The result is consistent in direction across all six seeds, but the effect is modest in magnitude. We therefore treat it as mechanistic supporting evidence rather than as a central claim.

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